

Finite Gamma coefficients, boundary resolvents, and consecutive quotient volumes

13 September 2026

This continuation proves a finite coefficient-to-metric map, its exact boundary-resolvent remainder, and a two-degree determinant formula for the consecutive volume ratio in the arithmetic Toda control. The original Gamma mass, tensor sum, arithmetic amplitude, quotient coordinates and relation polynomial remain in every formula. A bounded positive-weight calibration at the end proves that the relative Gamma correction can increase with degree even while the actual minimum volume decreases. That calibration is explicitly an auxiliary measure, with no assertion that its polynomial is an annihilator of actual zeta zeros.

The sources read for this calculation are the complete `RESEARCH_NOTE.md` and `NOTE.tex` in the delivered Tau Gamma Convolution Descent, and the original arithmetic definitions and proofs TVB.1–TVB.38 in `toda_cv_exact_bridge_20260912.tex`. The finite transfer below is at the original observation $\theta = 0$. Its proof does not replace a nonzero exponential tilt by a polynomial test. All interval claims here are finite rational inequalities; no uniform estimate as packet, tensor degree or polynomial degree grow is asserted.

1 The original source, relation and coordinate maps

Let $h(s)$ be a monic nonempty packet of actual zeros of

$$g(s) = s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) = 2\xi(s),$$

including the complete order at each selected zero. Put $v_h = g/h$, $w_h(t) = |v_h(1/2 + it)|^2/(2\pi)$, and $\mu_h = \int_{\mathbb{R}} w_h(t) dt$. The quotient is entire: local Taylor division removes exactly the stated vanishing order. Its value at each selected zero is nonzero. The full Taylor remainder $[v_h]_h$ is therefore a unit of $E_h = \mathbb{C}[s]/(h)$. Indeed in $\mathbb{C}[t]/(t^m)$ the inverse of $\sum_{j < m} a_j t^j$, $a_0 \neq 0$, has coefficients $b_0 = a_0^{-1}$ and $b_n = -a_0^{-1} \sum_{j=1}^n a_j b_{n-j}$; the Chinese remainder map from coprime factors is the exact algebra isomorphism obtained by Bezout's identity. This retains every jet and multiplicity.

The original tensor coordinate is

$$k \geq 1, \quad c = k/2, \quad S = s_1 + \cdots + s_k = c + iu, \quad u = t_1 + \cdots + t_k. \quad (\text{GMT.1})$$

The source observation is the conjugate-linear-first pairing

$$\langle P, Q \rangle_h = \int_{\mathbb{R}} \overline{P(c + iu)} Q(c + iu) m_{h,k}(u) du, \quad m_{h,k} = w_h^{*k}, \quad \int m_{h,k} = \mu_h^k. \quad (\text{GMT.2})$$

The change $(t_1, \dots, t_k) \mapsto (t_1, \dots, t_{k-1}, u)$ has determinant one. Thus Fubini gives (GMT.2) from the literal tensor Mellin observation without a divisor or an average coordinate. The complex Mellin amplitude remains

$$\prod_{i=1}^k \frac{v_h(1/2 + it_i)}{\sqrt{2\pi}} P(c + i \sum_i t_i).$$

Let $I = (h(s_1), \dots, h(s_k))$ and let χ be the monic generator of $I \cap \mathbb{C}[S]$, of degree q . Existence follows from an annihilating polynomial of multiplication by S on the finite algebra $\mathbb{C}[s_1, \dots, s_k]/I$. Division by its least-degree monic member proves that every member of the intersection is a multiple of χ . For $N \geq q - 1$ define $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$ and the exact maps

$$0 \longrightarrow \mathcal{P}_{N-q} \xrightarrow{M_\chi} \mathcal{P}_N \xrightarrow{J_N} \mathcal{C} \longrightarrow 0, \quad M_\chi P = \chi P, \quad \mathcal{C} = \mathbb{C}[S]/(\chi), \quad J_N P = [P]_\chi. \quad (\text{GMT.3})$$

Here $\mathcal{P}_{-1} = 0$. All coordinate vectors of $\mathcal{C}$ are in the written basis $1, S, \dots, S^{q-1}$. Its map to the original tensor jet is

$$\eta[P] = [v_h]_h^{\otimes k} [P(S)]_I.$$

Its kernel is zero by the preceding intersection ideal and invertible unit. No Gamma coefficient calculation changes this map.

Write B_N for the coefficient matrix of the columns $\chi, \chi S, \dots, \chi S^{N-q}$ in $\mathcal{P}_N$. The matrix $[E_N, B_N]$, where E_N includes the first q monomials, is triangular with diagonal one. Thus its determinant is exactly one in this order; putting the relation block first contributes $(-1)^{q(N-q+1)}$.

The original theta map is $\Theta\phi(x) = \sum_{n \neq 0} \phi(nx)$, with $D = -x\partial_x$, and F_h satisfies $\mathcal{M}F_h = v_h$, $h(D)F_h = \Theta\phi_0$, where $\phi_0 = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}$. The exact primitive attached to a column χP is retained as follows. Successive monic division in $s_1, \dots, s_k$ gives $\chi(S) = \sum_i h(s_i)Q_i(\mathbf{s})$. Its tensor primitive is

$$\sum_{i=1}^k (-1)^{i-1} Q_i(\mathbf{D}) P(\sum_j D_j) (F_h^{\otimes(i-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-i)}). \quad (\text{GMT.4})$$

The differential at the i th position has sign $(-1)^{i-1}$, so differentiation gives precisely the original source vector $\mathcal{V}_{h,k}(\chi P)$. This proves the relation-to-boundary map, including its sign. The construction of F_h uses the actual zero-division operator $S_\rho F(x) = x^{-\rho} \int_x^\infty F(y) y^{\rho-1} dy = -x^{-\rho} \int_0^x F(y) y^{\rho-1} dy$ at a vanishing Mellin value. The two integrals give respectively every large- and small- x power bound. Differentiation gives $(D - \rho)S_\rho F = F$, hence all Euler derivative bounds and the Mellin quotient by $s - \rho$. Starting from $\Theta\phi_0$ and successively dividing the complete root list constructs the displayed F_h . The seed identity follows by substituting $y = \pi x^2$ in its Mellin integral, which gives $\mathcal{M}\phi_0(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)$ for $\Re s > -2$, followed by the absolutely convergent theta sum for $\Re s > 1$ and analytic continuation.

On each original support label ℓ , a boundary still maps to $(\ell, 0) = e_\ell$; external absence τ still maps to τ . The linear polynomial maps (GMT.3)–(GMT.4) precede the quadratic map $R \mapsto R^* H R$ from source representatives to Hermitian forms. For the empty packet $h = 1$, the analytic source remains, but $\chi = 1$, $q = 0$, $\mathcal{C} = 0$, G_N is the zero-dimensional form and its determinant is one. Every quotient formula below then has its empty-block interpretation.

2 Full Gamma mass and the finite coefficient evaluation

For $\lambda > 0$ set

$$\alpha = 2\lambda, \quad \beta = k\alpha, \quad r_\lambda(t) = \frac{|\Gamma(\lambda + it/2)|^2}{2\pi}, \quad c_\lambda = 2^{1-2\lambda}\Gamma(2\lambda), \quad C_k = c_\lambda^k. \quad (\text{GMT.5})$$

The beta integral with its coordinate $x = 2y$ gives

$$|\Gamma(\lambda + it/2)|^2 = 2\Gamma(2\lambda) \int_{\mathbb{R}} e^{ity} (2 \cosh y)^{-2\lambda} dy.$$

Fourier inversion therefore gives $\int e^{iyt} r_\lambda(t) dt = c_\lambda (\cosh y)^{-2\lambda}$. Both functions are integrable: the right integrand and all its derivatives decay exponentially, so integration by parts bounds its Fourier transform by every inverse power. Exponential integrability of r_λ on $|a| < \pi/2$ follows

by rotating the gamma integral by any φ with $|a| < \varphi < \pi/2$; the squared modulus has a bound $\Gamma(\lambda)^2(\cos \varphi)^{-2\lambda}e^{-\varphi|t|}$. Analytic continuation now proves

$$\int e^{au} r_{\lambda,k}(u) du = C_k(\cos a)^{-\beta}, \quad r_{\lambda,k} = r_{\lambda}^{*k} = \frac{C_k}{c_{k\lambda}} r_{k\lambda}. \quad (\text{GMT.6})$$

The last identity is Fourier uniqueness with the exact product mass.

Let $b_j^{(\lambda)}$ denote the monic polynomials with

$$\sum_{j \geq 0} \frac{b_j^{(\lambda)}(t)}{j!} z^j = (1 - iz)^{-\lambda + it/2} (1 + iz)^{-\lambda - it/2} = (1 + z^2)^{-\lambda} e^{t \arctan z}. \quad (\text{GMT.7})$$

These agree with $j!P_j^{(\lambda)}(t/2; \pi/2)$ in the classical generating function at DLMF 18.23.7. The next proof fixes their scale without relying on that identification. Differentiating (GMT.7) gives

$$b_{j+1}^{(\lambda)} = t b_j^{(\lambda)} - j(j + \alpha - 1) b_{j-1}^{(\lambda)}.$$

For small real z, w , integration of the product of generating functions by (GMT.6) gives $c_\lambda(1 - zw)^{-\alpha}$, since $\cos(\arctan z + \arctan w) = (1 - zw)/\sqrt{(1 + z^2)(1 + w^2)}$. Local exponential domination justifies every coefficient derivative. Thus

$$\int b_i^{(\lambda)} b_j^{(\lambda)} r_\lambda = \delta_{ij} c_\lambda j! (\alpha)_j, \quad \int b_i^{(k\lambda)} b_j^{(k\lambda)} r_{\lambda,k} = \delta_{ij} C_k j! (\beta)_j. \quad (\text{GMT.8})$$

Put $B(t) = |v_h(1/2 + it)/\Gamma(\lambda + it/2)|^2$, so that $w_h = Br_\lambda$ with its full complex amplitude still present in the Mellin source. One may use the bounded reference constructed in the Gamma descent: $\lambda = 1/4$ for degree at least three, or $\lambda = 1/4 + L$, $L \geq \max(0, 3 - \deg h)$ otherwise. The relevant boundedness follows directly from the compact entire quotient, $|h(1/2 + it)| \geq (|t|/2)^{\deg h}$ beyond twice the root radius, $|\zeta(1/2 + it)| \leq 2(1 + |t|)$, and the nonvanishing factors $\prod_{j < L} (j + 1/4 + it/2)$ in the gamma shift. In particular $B \in L^2(r_\lambda dt)$. All statements of this section also hold for any specified nonnegative nonzero $B \in L^2(r_\lambda dt)$. For that domain write $w_B = Br_\lambda$ and $m_{B,k} = w_B^{*k}$; at the arithmetic input these equal the original $w_h, m_{h,k}$. Cauchy–Schwarz with (GMT.6) gives all polynomial moments. The calibration below uses this latter domain before it is explicitly capped to a bounded function.

The sum adjoint is the actual conditional integral

$$C_\Sigma F(u) = \frac{1}{r_{\lambda,k}(u)} \int_{\mathbb{R}^{k-1}} F(t_1, \dots, t_{k-1}, u - \sum_{i < k} t_i) \prod_{i < k} r_\lambda(t_i) r_\lambda(u - \sum_{i < k} t_i) d^{k-1}t. \quad (\text{GMT.9})$$

Its adjoint inclusion $U_\Sigma f(\mathbf{t}) = f(\sum_i t_i)$ is isometric by Fubini; hence $C_\Sigma U_\Sigma = 1$ and $F \mapsto (C_\Sigma F, (1 - U_\Sigma C_\Sigma)F)$ has inverse $(f, n) \mapsto U_\Sigma f + n$ on $L^2(r_{\lambda,k}) \oplus \ker C_\Sigma$. At $k = 1$ both maps are the identity. The normal component and every support label remain in this paired map. The specified sum density is $m_{B,k} = r_{\lambda,k} B_k$, $B_k = C_\Sigma B^{\otimes k}$; in particular it is $m_{h,k}$ at the arithmetic input.

Define the real coefficients $c_{B,j}$ (the original $c_{h,j}$ for the arithmetic multiplier $B = B_{h,\lambda}$) and formal coefficient power

$$c_{B,j} = \frac{\int b_j^{(\lambda)} w_B}{c_\lambda j! (\alpha)_j}, \quad A(z) = \sum_{j \geq 0} (\alpha)_j c_{B,j} z^j, \quad d_n = [z^n] A(z)^k. \quad (\text{GMT.10})$$

Multiplying the finite generating coefficients in (GMT.7) gives

$$b_n^{(k\lambda)}(\sum_i t_i) = \sum_{n_1 + \dots + n_k = n} \frac{n!}{\prod_i n_i!} \prod_i b_{n_i}^{(\lambda)}(t_i).$$

Pairing with $B^{\otimes k}$ gives the exact finite identity

$$\int b_n^{(k\lambda)}(u) m_{B,k}(u) du = C_k n! d_n. \quad (\text{GMT.11})$$

Every term is integrable by Cauchy–Schwarz; no infinite formal-series product was interchanged with an integral.

For a specified finite list of polynomials $p_0, \dots, p_r$ in $\mathcal{P}_N$, expand the actual tests

$$\overline{p_a(c + iu)} p_b(c + iu) = \sum_{n=0}^{2N} L_{ab,n} b_n^{(k\lambda)}(u).$$

This expansion is a finite algorithm: subtract the top coefficient times the monic polynomial of the same degree, then continue in descending degree through zero. The resulting remainder is exactly zero, by the decreasing degree. Equations (GMT.10)–(GMT.11) prove

$$\boxed{H(p)_{ab} = C_k \sum_{n=0}^{2N} n! L_{ab,n} d_n.} \quad (\text{GMT.12})$$

The coefficient-to-form map is homogeneous of degree k in $c_{B,0}, \dots, c_{B,2N}$. Its full coordinate type is $\mathbb{R}^{2N+1} \rightarrow \text{Herm}(\mathcal{P}_N)$ on the actual moment vectors, followed by congruence $H \mapsto P^* H P$ for a polynomial coefficient matrix P . The nonnegative density has positive mass. Its positivity set has positive Lebesgue measure, and a nonzero polynomial has only finitely many real roots. Thus the integral of its squared modulus is strictly positive, proving positivity of every finite source Gram and of every injective relation Gram. Choosing $p_a = S^a$ gives H_N ; choosing the columns of B_N gives exactly $B_N^* H_N B_N$. Choosing actual representatives gives their actual Gram. Every relation column has degree at most N , including its complete χ factor.

One may instead evaluate in the monic Gamma columns $p_j^\Gamma(S) = i^j b_j^{(k\lambda)}((S - c)/i)$. Their coefficient matrix U_N is triangular with diagonal one, and their reference Gram is

$$U_N^* H_N^\Gamma U_N = \text{diag}\{C_k j! (\beta)_j : 0 \leq j \leq N\}. \quad (\text{GMT.13})$$

This is a stated invertible coordinate map, with all squared norms retained. In particular it divides neither the measure nor the polynomials by their mass or norms. The original reduction matrix becomes $J_N U_N$; the original source form is recovered by $U_N^{-*}(\cdot) U_N^{-1}$.

3 Finite coefficient errors and polynomial-test bounds

Finite errors propagate before any inverse is taken. For real central coefficients $\widehat{c}_{B,j}$ and radii $e_j \geq 0$ enclosing the actual $c_{B,j}$ for $0 \leq j \leq 2N$, define the finite polynomials

$$\widehat{A} = \sum_{j=0}^{2N} (\alpha)_j \widehat{c}_{B,j} z^j, \quad A_+ = \sum_{j=0}^{2N} (\alpha)_j |\widehat{c}_{B,j}| z^j, \quad E_+ = \sum_{j=0}^{2N} (\alpha)_j e_j z^j,$$

and set

$$\widehat{d}_n = [z^n] \widehat{A}^k, \quad e_n^{(k)} = [z^n] \{(A_+ + E_+)^k - A_+^k\}. \quad (\text{GMT.14})$$

The multinomial expansion of each difference of products groups terms by the nonempty set of factors containing a coefficient error. Taking absolute values term by term proves $|d_n - \widehat{d}_n| \leq e_n^{(k)}$ for $n \leq 2N$. Thus the Hermitian central matrix $\widehat{H}$ from (GMT.12) satisfies

$$|(H - \widehat{H})_{ab}| \leq \varepsilon_{ab} := C_k \sum_{n=0}^{2N} n! |L_{ab,n}| e_n^{(k)}. \quad (\text{GMT.15})$$

The masses in this formula are exact constants. An interval evaluation of those constants is propagated by the same product rule; it cannot be replaced by setting $C_k = 1$. Exact polynomial coefficients of χ remain fixed in $L_{ab,n}$; any numerical enclosure of such coefficients must likewise be propagated before reporting a matrix certificate.

There is also a strictly polynomial-test refinement of the L^2 tail bound in the supplied descent. Write $B_L = \sum_{j \leq L} c_{B,j} b_j^{(\lambda)}$ and $E_k = B_k - C_\Sigma(B_L^{\otimes k})$. Orthogonality proves $\|B_L\| \leq \|B\|$. Expanding the tensor difference into k terms and using contraction of C_Σ proves $\|E_k\| \leq k\|B\|^{k-1}\|B - B_L\|$ with the original measures. Equation (GMT.11) applied degree by degree shows that E_k is orthogonal to all polynomials of degree at most L . Therefore

$$\left| \int \overline{p_a(c+iu)} p_b(c+iu) E_k(u) r_{\lambda,k}(u) du \right| \leq \|E_k\| \left(C_k \sum_{n > L} n! (\beta)_n |L_{ab,n}|^2 \right)^{1/2}. \quad (\text{GMT.16})$$

This follows by discarding only the orthogonal test components whose pairings are proved zero, and then applying Cauchy–Schwarz and (GMT.8). It is exactly zero at $L \geq 2N$. In particular a signed truncated density has the same finite positive Gram at that degree; pointwise positivity of the truncation is not required by this map.

An intrinsic finite polynomial-test bound is also available. The reference evaluation kernel is

$$\mathcal{K}_N^\Gamma(u) = \sum_{j=0}^N \frac{b_j^{(k\lambda)}(u)^2}{C_k j! (\beta)_j}.$$

Expansion in the original monic basis and Cauchy–Schwarz give $|P(c+iu)|^2 \leq \|P\|_\Gamma^2 \mathcal{K}_N^\Gamma(u)$. It follows that the form induced by a real error E_k obeys

$$|\langle P, P \rangle_{E_k}| \leq \|P\|_\Gamma^2 \|E_k\|_{L^2(r_{\lambda,k})} \|\mathcal{K}_N^\Gamma\|_{L^2(r_{\lambda,k})}. \quad (\text{GMT.17})$$

The last squared norm is an exact Gamma integral of a degree- $4N$ polynomial, evaluable by (GMT.8) or (GMT.12) with the reference weight. This is a finite evaluation-functional estimate, retaining every Christoffel-kernel denominator and the full mass.

4 Exact least lifts and the boundary-resolvent remainder

For a positive definite source matrix H in the fixed source coordinates, define

$$K(H) = J_N H^{-1} J_N^*, \quad G(H) = K(H)^{-1}, \quad R(H) = H^{-1} J_N^* G(H). \quad (\text{GMT.18})$$

Surjectivity of J_N makes $K(H)$ positive definite. Multiplication gives $J_N R(H) = 1$ and $B_N^* H R(H) = 0$. Every other lift is $R(H)x + B_N y$, whose squared norm is $x^* G(H)x + y^* B_N^* H B_N y$. This proves minimum, uniqueness and $R(H)^* H R(H) = G(H)$.

Let H_0 and H_1 be two actual positive finite source forms on the same polynomial space. In particular they may be the Gamma and arithmetic forms already constructed. Put

$$H_t = H_0 + tE, \quad E = H_1 - H_0, \quad 0 \leq t \leq 1, \quad A_0 = B_N^* H_0 B_N, \quad F = B_N^* E B_N, \quad C = B_N^* E R_0,$$

where $R_0 = R(H_0)$ and $G_0 = G(H_0)$. The whole interval of forms is positive because it is the convex combination of the two actual positive forms. Thus every nonempty $A_t = A_0 + tF$ is positive. The complete, signed resolvent formulas are

$$\boxed{R_t = R_0 - t B_N A_t^{-1} C, \quad G_t = G_0 + t R_0^* E R_0 - t^2 C^* A_t^{-1} C.} \quad (\text{GMT.19})$$

To prove the first equation, its reduction is the identity and $B_N^* H_t R_0 = tC$, while $B_N^* H_t B_N = A_t$. Consequently its boundary pairing is zero, so (GMT.18) identifies it. Expanding its full H_t norm

gives $R_0^* H_t R_0 - 2t^2 C^* A_t^{-1} C + t^2 C^* A_t^{-1} A_t A_t^{-1} C$. The two negative terms and one positive term yield exactly (GMT.19). In particular the unscaled matrix $C^* A_t^{-1} C$ is positive semidefinite and vanishes precisely when $C = 0$. Pairing $A_t^{-1} > 0$ with every column of C proves this criterion. The full quadratic loss $t^2 C^* A_t^{-1} C$ vanishes precisely when $t = 0$ or $C = 0$; for an empty boundary block it is the zero form throughout.

The representative difference is a literal boundary with coefficient vector $-t A_t^{-1} C x$; inserting that polynomial into (GMT.4) gives its full original theta primitive. The same jets ηx persist. No statement of metric nonidentity leaves its comparison map unspecified.

For completeness the exact derivative identities are

$$R'_t = -B_N A_t^{-1} B_N^* E R_t, \quad G'_t = R_t^* E R_t, \quad G''_t = -2(B_N^* E R_t)^* A_t^{-1} (B_N^* E R_t). \quad (\text{GMT.20})$$

Differentiate $J_N R_t = 1$ and $B_N^* H_t R_t = 0$ to obtain the first formula. The derivative R'_t is a boundary, hence its two pairings with R_t in the derivative of $R_t^* H_t R_t$ vanish. This proves the second formula. Differentiating it and substituting the first proves the third. This is a proved concavity statement along the actual finite metric segment, with the entire boundary coupling retained.

The determinant-line calculation has no hidden source-size exponent. The change $[E_N, B_N] \mapsto [R(H), B_N]$ adds boundary columns to the first q columns and has determinant one. Its Gram is block diagonal. Therefore

$$V_N(H) = \det G(H) = \frac{\det H}{\det(B_N^* H B_N)}. \quad (\text{GMT.21})$$

For the original arithmetic and Gamma forms this is exactly $V_N = V_N^\Gamma T_N$, with the same $T_N = X_{N+1}/Y_{N-q+1}$ as the supplied descent. Formula (GMT.19) gives its complete matrix-level map and its nonlinear remainder.

5 A two-degree map for the actual consecutive ratio

Fix $N \geq q$. In original source monomials split degree $N + 1$ into degrees $0, \dots, N - 1$ followed by S^N, S^{N+1} , and write

$$H_+ = \begin{pmatrix} H_- & C_D \\ C_D^* & D_D \end{pmatrix}, \quad W = D_D - C_D^* H_-^{-1} C_D, \quad J_+ = [J_-, J_D], \quad F_D = J_D - J_- H_-^{-1} C_D. \quad (\text{GMT.22})$$

Here $H_- = H_{N-1}$, $H_+ = H_{N+1}$, W is 2×2 , and $F_D : \mathbb{C}^2 \rightarrow \mathcal{C}$. Its columns are the quotient classes of the two specified polynomials after subtracting their actual orthogonal projections onto $\mathcal{P}_{N-1}$. The block map

$$(x, y) \mapsto (x - H_-^{-1} C_D y, y)$$

has determinant one and carries the source Gram to $\text{diag}(H_-, W)$. Positivity proves $W > 0$ and, by applying the reduction map to this exact orthogonal direct sum,

$$K_+ = K_- + F_D W^{-1} F_D^*, \quad Z_D = F_D^* G_- F_D \succeq 0, \quad G_- = K_-^{-1}. \quad (\text{GMT.23})$$

Block elimination of $\begin{pmatrix} I & A \\ -B & I \end{pmatrix}$ in its two orders proves $\det(I + AB) = \det(I + BA)$. Applying this identity to (GMT.23) gives the exact scalar maps

$$\boxed{\frac{V_{N-1}}{V_{N+1}} = \frac{\det(W + Z_D)}{\det W} = 1 + \text{Tr}(W^{-1} Z_D) + \frac{\det Z_D}{\det W}.} \quad (\text{GMT.24})$$

The last equality is the complete characteristic polynomial of a 2×2 matrix. All denominators are positive. This gives the actual monotonicity because both eigenvalues of $W^{-1/2} Z_D W^{-1/2}$ are nonnegative.

The same block also gives the adjacent source recurrence coefficient. In the order in (GMT.22), the monic residual of S^N has squared norm W_{11} , and eliminating it from the residual of S^{N+1} gives squared norm $\det W/W_{11}$. Therefore

$$\boxed{a_{N+1} = \frac{\omega_{N+1}}{\omega_N} = \frac{\det W}{W_{11}^2}.} \quad (\text{GMT.25})$$

The first residual is monic because the subtracted projection has smaller degree; so is the second after its additional elimination. The norms have not been rescaled. The coefficient map $S^j = \sum_{l \leq j} \binom{j}{l} c^{j-l} i^l u^l$ has determinant $i^{j(j-1)/2}$ in dimension j and hence Gram determinant multiplier one, which proves that these are also the source Hankel–Toda norms.

Apply (GMT.22)–(GMT.25) separately to the Gamma reference and the arithmetic source, retaining their same J_+ and χ . Then

$$\boxed{\frac{T_{N-1}}{T_{N+1}} = \frac{\det(W + Z_D)/\det W}{\det(W^\Gamma + Z_D^\Gamma)/\det W^\Gamma}.} \quad (\text{GMT.26})$$

Every entry is a finite polynomial in the one-factor coefficients through degree $2N + 2$, followed by explicitly stated inverses of positive matrices. In the supplied Toda upper estimate, substitute the two positive scalars $a_{N+1} = \det W/W_{11}^2$ and $\mathcal{R}_N = \det(W + Z_D)/\det W$. Its full right side becomes

$$\sqrt{\frac{\det W}{W_{11}^2} \frac{\det(W + Z_D)/\det W - 1}{2\sqrt{\det(W + Z_D)/\det W}}}. \quad (\text{GMT.27})$$

This is an exact substitution into that existing inequality; its discarded phase and volume-imbalance squares remain the ones proved in TVB.32–TVB.34a. Formula (GMT.27) alone does not remove them or assert equality. The two-degree determinant formula itself also works at $q = 0$, where F_D and Z_D are zero and the ratio is one.

6 Certified finite envelopes and quotient-dimensional bounds

Suppose a finite coefficient calculation supplies the entry bounds (GMT.15), or the proved polynomial-test bounds (GMT.16). Let $v_i > 0$ be any specified positive rational weights and set

$$\Delta = \text{diag}(\delta_i), \quad \delta_i = \sum_j \varepsilon_{ij} \frac{v_j}{v_i}, \quad H^{\text{lo}} = \widehat{H} - \Delta, \quad H^{\text{hi}} = \widehat{H} + \Delta. \quad (\text{GMT.28})$$

The radii may be replaced by symmetric rational upper radii. For each pair i, j , the inequality $2|x_i||x_j| \leq (v_j/v_i)|x_i|^2 + (v_i/v_j)|x_j|^2$ gives $|x^*(H - \widehat{H})x| \leq x^*\Delta x$ after summation. This proves $H^{\text{lo}} \preceq H \preceq H^{\text{hi}}$ without using a pointwise lower bound on B .

The positive-pivot certificate for H^{lo} is entirely finite. Eliminate successively its first diagonal pivot, recording

$$\begin{pmatrix} a & b^* \\ b & C \end{pmatrix} \mapsto \text{diag}(a, C - bb^*/a), \quad (x, y) \mapsto (x - a^{-1}b^*y, y).$$

The displayed congruence is invertible. Positive recorded rational pivots prove positivity by induction; nonpositive pivots reject that particular enclosure. Actual matrix positivity is already proved by the source integral. For any sequence of valid entry intervals tending to their actual entries, the lower envelopes tend to that positive matrix. Its least unit-sphere quadratic value is positive by compactness, and convergence in the finite sum of entry errors eventually makes every lower envelope positive. Hence the certificate process succeeds at every fixed finite degree when its actual input intervals are refined; no success or interval refinement is asserted for absent arithmetic data.

For a positive certified lower envelope, calculate the actual finite quotient forms $G^{\text{lo}}, G^{\text{hi}}$ by (GMT.18), or by the equivalent Schur formula

$$G(H) = E_N^* H E_N - E_N^* H B_N (B_N^* H B_N)^{-1} B_N^* H E_N. \quad (\text{GMT.29})$$

The formula follows by substituting $E_N - B_N (B_N^* H B_N)^{-1} B_N^* H E_N$ for the unique least lift. The same radii certify the representative coefficients themselves. Write $\hat{R} = R(\hat{H})$, $A^{\text{lo}} = B_N^* H^{\text{lo}} B_N$ and define, using any rational upper bounds for the displayed complex moduli,

$$\begin{aligned} s_{ai} &= \sum_{r,l} |(B_N)_{ra}| \varepsilon_{rl} |\hat{R}_{li}|, \\ k_{ab} &= \frac{1}{2} ((A^{\text{lo}})_{aa}^{-1} + (A^{\text{lo}})_{bb}^{-1}), \\ e_{ri}^R &= \sum_{a,b} |(B_N)_{ra}| k_{ab} s_{bi}. \end{aligned}$$

Then the full original-coordinate error and its quadratic loss obey

$$\boxed{|(R(H) - \hat{R})_{ri}| \leq e_{ri}^R, \quad \hat{R}^* H \hat{R} - G(H) = (R(H) - \hat{R})^* H (R(H) - \hat{R}).} \quad (\text{GMT.29a})$$

To prove the entry bound set $A = B_N^* H B_N$ and $S_E = B_N^* (H - \hat{H}) \hat{R}$. Formula (GMT.19) gives $R(H) - \hat{R} = -B_N A^{-1} S_E$. The triangle inequality gives $|(S_E)_{ai}| \leq s_{ai}$. Inversion reverses positive order: conjugating $A \succeq A^{\text{lo}}$ by $(A^{\text{lo}})^{-1/2}$, inverting its eigenvalues, and conjugating back gives $A^{-1} \preceq (A^{\text{lo}})^{-1}$. Positivity of each two-coordinate Gram minor of A^{-1} gives $|A_{ab}^{-1}| \leq \sqrt{A_{aa}^{-1} A_{bb}^{-1}} \leq k_{ab}$, where the last step is the arithmetic-geometric mean inequality and the proved diagonal bounds. Matrix multiplication now proves the coefficientwise bound in (GMT.29a). For its second identity, $\hat{R} - R(H)$ is an actual boundary, orthogonal to $R(H)$ in H . Expanding $\hat{R}^* H \hat{R}$ proves the equality.

This loss has an explicit q -dimensional rational upper form. Choose positive rational weights v_i in the original quotient coordinates and set

$$\tau_{ij} = \sum_{a,b} s_{ai} |(A^{\text{lo}})_{ab}^{-1}| s_{bj}, \quad \Delta_Q = \text{diag} \left\{ \sum_j \tau_{ij} \frac{v_j}{v_i} \right\}.$$

The same entrywise quadratic inequality as (GMT.28) proves

$$\boxed{0 \leq \hat{R}^* H \hat{R} - G(H) \leq \Delta_Q.} \quad (\text{GMT.29b})$$

Indeed the middle matrix is $S_E^* A^{-1} S_E \preceq S_E^* (A^{\text{lo}})^{-1} S_E$. Its value on x is at most $\sum_{i,j} \tau_{ij} |x_i| |x_j| \leq x^* \Delta_Q x$. Every source row, relation coefficient and quotient coordinate is retained in these sums. At an empty boundary block both error forms are zero. Applying $\mathcal{V}_{h,k}$ to each representative difference gives the theta primitive (GMT.4), so the certified coefficient error also has its original source and quotient interpretation.

Minimizing the two proven source inequalities over the same affine space of representatives gives

$$G^{\text{lo}} \preceq G(H) \preceq G^{\text{hi}}, \quad \det G^{\text{lo}} \leq V_N(H) \leq \det G^{\text{hi}}. \quad (\text{GMT.30})$$

For the determinant assertion, conjugate by the positive square root of the lower form and multiply the eigenvalues, all of which are at least one; apply the same argument for the upper form.

Construct the degree $N - 1$ envelope by taking the corresponding principal submatrix of the same degree $N + 1$ envelope. Then

$$\frac{\det G_-^{\text{lo}}}{\det G_+^{\text{hi}}} \leq \frac{V_{N-1}}{V_{N+1}} \leq \frac{\det G_-^{\text{hi}}}{\det G_+^{\text{lo}}}. \quad (\text{GMT.31})$$

Division by the exact positive Gamma ratio from (GMT.26) gives the certified interval for T_{N-1}/T_{N+1} . These are determinants of size q , with no independent numerator/denominator error estimate on the larger source and relation determinants.

For example a relative certificate $(1-\eta)\widehat{H} \preceq H \preceq (1+\eta)\widehat{H}$, with a recorded rational $0 \leq \eta < 1$ and $\widehat{H} > 0$, gives exactly

$$(1-\eta)^q \det G(\widehat{H}) \leq V_N(H) \leq (1+\eta)^q \det G(\widehat{H}),$$

and a common such certificate for the two nested degrees gives ratio multipliers $((1-\eta)/(1+\eta))^q$ and its reciprocal. This follows because $R(aH) = R(H)$ and $G(aH) = aG(H)$ for $a > 0$, as direct substitution in (GMT.18) proves. The mass is scaled only in this explicitly declared comparison, never assigned a new value in the original arithmetic source.

7 Exact calibration and a bounded relative-volume counterexample

Take the explicitly auxiliary one-factor weight

$$\lambda = 1, \quad k = 2, \quad c = 1, \quad c_1 = \frac{1}{2}, \quad C_2 = \frac{1}{4}, \quad B(t) = 1 + t^2/4, \quad w = Br_1, \quad \chi(S) = (S-1)^2 - 2. \quad (\text{GMT.32})$$

The positive polynomial B lies in $L^2(r_1)$ by (GMT.6). It is not the function $|g/h\Gamma|^2$ of a claimed zeta packet. Since $b_2^{(1)} = t^2 - 2$, its coefficients are $c_{B,0} = 3/2$, $c_{B,2} = 1/4$ and all others zero. Therefore

$$A(z) = \frac{3}{2} + \frac{3}{2}z^2, \quad (d_0, d_2, d_4) = (\frac{9}{4}, \frac{9}{2}, \frac{9}{4}), \quad B_2(u) = \frac{54}{35} + \frac{39}{280}u^2 + \frac{3}{1120}u^4. \quad (\text{GMT.33})$$

To verify the last equality, use $b_2^{(2)} = u^2 - 4$, $b_4^{(2)} = u^4 - 32u^2 + 72$ in $\sum_n d_n b_n^{(2)}/(4)_n$ and expand all three terms. The one-factor mass is $3/4$ and the sum mass is $9/16$. Repeated multiplication by t in the recurrence (GMT.8) gives the one-factor reference even moments through degree eight as $(1/2, 1, 8, 136, 3968)$. Inserting $B = 1 + t^2/4$ and expanding $(t_1 + t_2)^n$ with the full binomial coefficients gives sum moments

$$(\mu_0, \mu_1, \dots, \mu_8) = (9/16, 0, 9/2, 0, 117, 0, 5472, 0, 385272). \quad (\text{GMT.34})$$

This is also (GMT.11)–(GMT.12); the two calculations independently fix its mass and tensor cross term.

The original degree-three source Gram and reference Gram are

$$H_3 = \frac{1}{16} \begin{pmatrix} 9 & 9 & -63 & -207 \\ 9 & 81 & 81 & -1863 \\ -63 & 81 & 2025 & 2025 \\ -207 & -1863 & 2025 & 93393 \end{pmatrix}, \quad H_3^\Gamma = \frac{1}{4} \begin{pmatrix} 1 & 1 & -3 & -11 \\ 1 & 5 & 5 & -55 \\ -3 & 5 & 65 & 65 \\ -11 & -55 & 65 & 1685 \end{pmatrix}. \quad (\text{GMT.35})$$

Every entry follows by expanding $(1 - iu)^a(1 + iu)^b$ against (GMT.34), or against the reference moments. The complete original relation and reduction matrices are

$$B_3 = \begin{pmatrix} -1 & 0 \\ -2 & -1 \\ 1 & -2 \\ 0 & 1 \end{pmatrix}, \quad J_3 = \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & 2 & 5 \end{pmatrix}. \quad (\text{GMT.36})$$

These follow from $S^2 = 2S + 1$ and $S^3 = 5S + 2$ modulo χ ; multiplication verifies $J_3 B_3 = 0$ and $\det[E_3, B_3] = 1$. Direct substitution into (GMT.18) gives

$$G_1 = \frac{1}{16} \begin{pmatrix} 9 & 9 \\ 9 & 81 \end{pmatrix}, \quad G_3 = \begin{pmatrix} 81/244 & 81/244 \\ 81/244 & 175041/80764 \end{pmatrix},$$

$$R_3 = \begin{pmatrix} 117/122 & -801/40382 \\ -5/61 & 18963/20191 \\ 5/122 & -907/40382 \\ 0 & 7/331 \end{pmatrix}, \quad G_3^\Gamma = \begin{pmatrix} 5/38 & 5/38 \\ 5/38 & 2255/4142 \end{pmatrix}. \quad (\text{GMT.37})$$

These values can be checked without trusting an inverse algorithm: multiply to obtain $J_3 R_3 = I$, $B_3^* H_3 R_3 = 0$ and $R_3^* H_3 R_3 = G_3$, then use the uniqueness proved in (GMT.18).

For the degree-one to degree-three split, the full two-degree data are

$$W = \begin{pmatrix} 81 & 243 \\ 243 & 3159 \end{pmatrix}, \quad F_D = \begin{pmatrix} 10 & 2 \\ 0 & 28 \end{pmatrix}, \quad Z_D = \frac{1}{4} \begin{pmatrix} 225 & 675 \\ 675 & 16137 \end{pmatrix}. \quad (\text{GMT.38})$$

Equation (GMT.22), evaluated with (GMT.35)–(GMT.36), proves these entries by ordinary rational multiplication. Consequently

$$\frac{V_1}{V_3} = \frac{20191}{4860}, \quad a_3 = 30, \quad \frac{V_1^\Gamma}{V_3^\Gamma} = \frac{2071}{450}, \quad \boxed{\frac{T_1}{T_3} = \frac{100955}{111834} < 1}. \quad (\text{GMT.39})$$

Here $V_1^\Gamma = 1/4$, $V_3^\Gamma = 225/4142$, obtained by taking the two displayed 2×2 determinants. Thus no unrecorded cancellation of reference masses is used.

The phenomenon has a completely specified bounded positive version. Define $B^{[200]}(t) = \min(1 + t^2/4, 10001)$ and $w^{[200]} = B^{[200]} r_1$. This changes the auxiliary weight only; it leaves $S = 1 + iu$, χ , the reference Gamma mass and the unscaled convolution operation fixed. For $0 \leq j \leq 6$,

$$\int |t|^j (w - w^{[200]})(t) dt \leq \frac{1}{4} \int_{|t| > 200} |t|^{j+2} r_1(t) dt < \frac{8!}{2^{92}} =: \epsilon_*. \quad (\text{GMT.40})$$

Here is a proof with no floating evaluation. For every integer $r \geq 0$, $x^r \leq r! 2^r e^{x/2}$; on $x > 200$ multiply by $e^{(x-200)/2} \geq 1$. Equation (GMT.6) and evenness give

$$\int_{|t| > 200} |t|^r r_1(t) dt \leq 2c_1 r! 2^r e^{-100} (\cos 1)^{-2} \leq 4r! 2^r e^{-100}.$$

The estimate $\cos 1 \geq 1/2$ follows from $1 - \cos 1 = \int_0^1 \sin x dx \leq \int_0^1 x dx = 1/2$, using $|\sin x| \leq x$ on this interval. Since $e > 2$, $e^{-100} < 2^{-100}$. Use $r = j + 2 \leq 8$ and the factor $1/4$ to obtain (GMT.40).

For $0 \leq j \leq 6$, the absolute j th one-factor moment of w is at most $A_* = \mu_0(w) + \mu_6(w) = 3/4 + 1128 = 4515/4$, because $|t|^j \leq 1 + |t|^6$. The exact tensor difference is $(w - w^{[200]}) \otimes w + w^{[200]} \otimes (w - w^{[200]})$. Expanding $|t_1 + t_2|^n \leq \sum_j \binom{n}{j} |t_1|^j |t_2|^{n-j}$ proves a sum-moment difference at degree $n \leq 6$ of at most $2^{n+1} A_* \epsilon_* \leq 128 A_* \epsilon_*$. The coefficient absolute sum in $(1 - iu)^a (1 + iu)^b$ is at most $2^{a+b} \leq 64$ for $a, b \leq 3$. Hence every source entry changes by at most $8192 A_* \epsilon_* < 1/40000$. The last inequality is an integer comparison after substitution of A_*, ϵ_* . Equation (GMT.28) with four weights equal to one now proves

$$H_3 - 10^{-4} I \prec H_3^{[200]} \prec H_3 + 10^{-4} I. \quad (\text{GMT.41})$$

Strictness follows because the error in each row sum is strictly less than 10^{-4} . The four leading principal minors of the lower matrix are

$$\frac{703}{1250}, \quad \frac{253068751}{100000000}, \quad \frac{102481960035937}{500000000000}, \quad \frac{4980299350501627807501}{10000000000000000}.$$

They are positive; their successive ratios are exactly the positive pivots in the elimination proving (GMT.28). Strict source inequalities give strict quotient inequalities: evaluate the lower form on the actual minimum lift, and the actual form on the upper minimum lift. Each lift of a nonzero

quotient vector is nonzero, so the positive source difference has a strictly positive quadratic value. Computing (GMT.29) for these two rational endpoint matrices, and the degree-one principal submatrices, gives the strict ratio enclosure

$$\frac{20694424798404596925036}{4984219548001747192501} < \frac{V_1^{[200]}}{V_3^{[200]}} < \frac{20703435010591353075036}{4980299350501627807501} < \frac{2071}{450}. \quad (\text{GMT.42})$$

The first two strict comparisons follow from that strict minimization and (GMT.31); the final comparison follows by positive-denominator cross multiplication. The rational envelope computation is reproduced by the exact checker accompanying this proof. Division by the fixed positive Gamma ratio therefore proves $T_1^{[200]}/T_3^{[200]} < 1$ for a function bounded between 1 and 10001. This is an explicit obstruction to deriving relative volume monotonicity from positivity and boundedness of a multiplier alone. Its exact relation to the arithmetic problem is the same coefficient, relation, least-lift and determinant maps (GMT.12), (GMT.18), (GMT.24) and (GMT.26). It does not assert that the actual arithmetic multiplier has this behavior.

8 Verification scope

The accompanying exact checker constructs moments in two ways: Jacobi recurrence and the literal two-factor binomial expansion, then compares them with the Gamma coefficient formula. It verifies the original S -coordinate Gram, every displayed lift and relation identity, the full boundary-resolvent loss, five points on its actual positive matrix segment, both two-degree formulas and the rational envelope calculation. Its deliberate formula changes remove the mass, omit the tensor cross term, reverse the boundary-loss sign, remove the conjugation, or remove one power in the Schur denominator. Each change is rejected by the corresponding exact check in ordinary and optimized Python. The written proofs establish the all-degree identities; these finite checks provide reproducible regression evidence. There is no Lean execution or analytic interval claim for a nonempty zeta packet in this verification.